

MATSUMOTO, Japan

WMO: 476180

Lat: 36.2467N

Lon: 137.9706E

Elev: 612

StdP: 94.19

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.8	-14.1	1.2	-1.1	-12.6	1.4	-1.6	8.0	6.4	7.3	5.5	1.4	10	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.9	33.4	22.7	32.0	22.2	30.5	21.7	23.7	30.6	23.2	29.6	22.6	28.5	3.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.9	26.2	21.3	17.2	25.9	20.8	16.7	25.6	74.6	30.8	72.3	30.0	70.1	28.5	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.7	6.0	DB	-11.0	35.3	1.3	1.1	-11.9	36.1	-12.7	36.7	-13.4	37.3	-14.4	38.1	(4)
			WB	-11.6	24.9	1.2	0.5	-12.5	25.2	-13.2	25.5	-13.9	25.8	-14.8	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-0.2	0.9	4.9	11.1	17.0	20.7	25.0	25.9	21.1	14.6	8.1	2.8	(6)
		DBStd	9.49	2.72	3.24	3.61	4.00	2.96	2.58	2.40	2.28	3.43	3.31	3.17	2.97	(7)
		HDD10.0	1071	315	256	163	34	1	0	0	0	3	74	224		(8)
		HDD18.3	2690	573	488	416	217	61	8	0	14	124	307	482		(9)
		CDD10.0	2063	0	1	6	68	217	322	463	492	332	145	18	1	(10)
		CDD18.3	642	0	0	0	1	19	80	205	233	96	8	0	0	(11)
		CDH23.3	4857	0	0	0	26	206	504	1597	1941	552	32	0	0	(12)
		CDH26.7	1691	0	0	0	3	38	104	602	793	148	2	0	0	(13)
(14)	Wind	WSAvg	2.4	2.2	2.3	2.7	2.8	2.7	2.3	2.4	2.3	2.1	2.0	2.2	2.3	(14)
(15)	Precipitation	PrecAvg	1024	34	41	71	81	96	139	135	93	148	100	52	27	(15)
(16)		PrecMax	1404	118	116	153	192	272	319	367	226	305	389	170	84	(16)
(17)		PrecMin	640	0	5	5	9	32	36	22	7	17	6	4	0	(17)
(18)		PrecStd	195	27	28	33	37	46	62	75	54	73	67	34	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	11.8	15.1	20.4	26.6	30.0	31.1	34.7	35.1	32.3	26.4	20.4	15.0		(19)
(20)		0.4% MCWB	6.6	8.2	9.5	13.3	16.0	19.2	23.0	23.1	21.6	18.1	12.5	9.5		(20)
(21)		2% DB	8.9	12.0	16.9	23.2	27.6	29.3	33.2	33.7	30.1	23.9	18.0	12.3		(21)
(22)		2% MCWB	4.1	6.4	8.2	12.1	15.6	19.1	22.8	22.8	21.3	16.8	11.5	7.4		(22)
(23)		5% DB	6.9	9.3	14.5	21.0	25.6	27.7	31.9	32.4	28.3	22.1	16.0	10.4		(23)
(24)		5% MCWB	2.7	4.3	7.4	11.4	14.6	18.5	22.3	22.5	20.6	16.1	10.8	6.0		(24)
(25)		10% DB	5.1	7.2	12.2	18.9	23.8	26.4	30.2	31.0	26.5	20.3	14.2	8.5		(25)
(26)		10% MCWB	1.3	2.8	6.5	10.8	14.2	18.3	21.9	22.4	20.0	15.0	9.9	4.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	8.0	10.0	11.9	16.7	19.2	22.2	24.6	24.6	23.0	19.7	14.7	11.2		(27)
(28)		0.4% MCDB	10.6	13.5	15.4	19.8	24.3	27.6	31.9	31.9	29.1	24.2	17.6	13.3		(28)
(29)		2% WB	4.8	7.0	10.0	14.3	17.8	21.2	23.8	23.9	22.1	18.3	13.1	8.4		(29)
(30)		2% MCDB	7.8	11.1	14.1	20.4	23.6	25.7	30.6	30.9	27.6	22.2	16.0	11.3		(30)
(31)		5% WB	2.9	4.9	8.5	12.9	16.5	20.3	23.1	23.4	21.5	17.1	11.8	6.4		(31)
(32)		5% MCDB	6.6	8.5	13.4	18.6	22.7	25.2	29.6	30.0	26.4	20.9	14.9	9.6		(32)
(33)		10% WB	1.5	3.1	6.8	11.6	15.5	19.5	22.5	22.9	20.8	15.8	10.3	4.8		(33)
(34)		10% MCDB	4.6	6.7	11.7	17.2	21.4	24.5	28.4	29.0	25.2	19.2	13.6	8.0		(34)
(35)	Mean Daily Temperature Range	MDBR	9.2	10.1	11.3	12.5	12.2	10.1	9.3	9.9	9.1	9.8	10.4	9.4		(35)
(36)		5% DB MCDBR	11.7	13.8	15.6	16.1	15.6	12.6	11.7	11.9	11.0	11.5	13.1	11.8		(36)
(37)		5% WB MCWBR	7.7	8.7	8.5	7.2	6.1	4.6	3.9	3.7	4.0	5.4	7.5	7.8		(37)
(38)		5% WB MCDBR	10.7	12.2	12.9	12.5	12.4	9.4	10.5	10.6	9.5	9.2	10.4	10.6		(38)
(39)		5% WB MCWBR	7.5	8.4	8.3	6.8	5.9	4.1	3.9	3.7	3.8	4.8	6.8	7.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.366	0.409	0.481	0.526	0.544	0.580	0.561	0.565	0.490	0.435	0.385	0.361		(40)
(41)		taud	2.044	1.946	1.843	1.834	1.845	1.822	1.941	1.922	2.078	2.145	2.212	2.162		(41)
(42)		Ebn at Noon	792	802	778	769	762	732	744	729	766	772	774	775		(42)
(43)		Edh at Noon	127	159	193	206	207	211	186	186	151	129	108	107		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.17	2.95	3.83	4.68	5.25	4.78	4.68	4.74	3.82	3.16	2.38	1.91		(44)
(45)		RadStd	0.16	0.19	0.26	0.39	0.55	0.34	0.68	0.47	0.32	0.30	0.19	0.18		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page